

ARNAB KUMAR DEY

arnabkumardey9957@gmail.com | (+91) 7407431911 | Kolkata, West Bengal, India
linkedin.com/in/arnab-kumardey | github.com/arnab9957 | leetcode.com/u/arnab9957

EDUCATION

Bachelor of Technology (BTech) in Artificial Intelligence

Alipurduar Government Engineering and Management College

Minor Degree in Artificial Intelligence

Indian Institute of Technology (IIT), Raipur (RPR)

Expected Graduation: 2028

YGPA: 8.19

Completed

Project Grade: 9.1

RESEARCH EXPERIENCE

Research Intern

Institute of Engineering & Management (IEM), Kolkata

Present

Advisor: Dr. Subhabrata Banerjee

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, C, C++, Dart

Frontend: React.js, Flutter, HTML, CSS, Tailwind, Bootstrap, Next.js

Backend & Databases: Node.js, Express.js, MySQL, Firebase, Inngest, Supabase

AI/ML: PyTorch, Neural Networks, NLP, RAG, LLMs (Basics), XAI

Tools & Platforms: Git, GitHub, Vercel, Render, Docker, Google Cloud, Jupyter

PROJECTS

- **RAG-ISRO:** a mission-critical, self-contained multi-agent RAG engine designed for secure, air-gapped environments to query aerospace technical specifications and compliance guidelines with built-in zero-trust security pipelines, dynamic access control, and formal verification.
- **Business Challenges AI-Powered Solutions:** Developed end-to-end AI models to solve real-world industry bottlenecks.
- **HealthWise:** A smart symptom detection system powered by ML to assist primary health analysis.
- **Zoom Clone:** Video conferencing application with individual and peer-to-peer connection capabilities.
- **OpenCV Projects:** Featuring an interactive AI story generator mixing vision processing and generation.
- **Roof-Top-Calc:** Automated utility tool designed for rooftop surface calculation via computer vision.
- **LLM Bias Mitigation:** A powerful tool to detect, analyze, and mitigate algorithmic bias in Large Language Models.
- **Narrative Consistency Reasoning System:** A sophisticated multi-stage AI pipeline that verifies whether hypothetical character backstories remain logically consistent.

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

- Active problem solver on **LeetCode**, **NeetCode**, and **GeeksforGeeks (GFG)**.
- **Core Competencies:** Advanced Dynamic Programming, Hash Tables, Greedy Algorithms, Complex Data Structures (Arrays, Strings).

ADDITIONAL INFORMATION

Languages: English (Fluent), Hindi (Fluent), Bengali (Native)

Currently Learning: Agentic AI, Advanced Machine Learning, Cloud Computing (AWS/Azure), DevOps Technologies

Summary: Aspiring Software & AI Engineer skilled in building scalable full-stack and AI-enabled solutions. Experienced in hands-on project implementation with a solid competitive programming background, seeking internship opportunities in AI/ML, Backend, or Software Engineering roles.